

A Software-Based Approach to Remote Motor and Cognitive Data Collection

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Abstract—Neurological degenerative diseases like Alzheimer’s Disease affect millions of individuals worldwide. While there is no cure for this disease, early detection can improve patient outcomes by enabling earlier intervention and treatment strategies. One method of detecting these early stages is to use the Kinesiological Instrument for Normal and Altered Reaching Movement (KINARM) lab, particularly its Object Hit and Avoid (OHA) task. This task specifically measures the subject’s rapid motor decisions, inhibitory control, and spatial attention, requiring them to use their arms to hit incoming objects while also avoiding “distractors.” Space Trash seeks to recreate the OHA task as a downloadable, gamified application to enable data collection outside the development stage of several methods of recreating this bimanual task, with our two versions being created with multi-mouse and camera tracking. In our multi-mouse implementation, the game is played by using two separate mice, controlling separate cursors on the screen. Even though multi-mouse is closer to its first debut, we’ve realized a flaw regarding the player playing on a different

plane than the screen. To combat this, we’ve begun developing a camera-tracking implementation of the task, allowing the user to play on the same plane as the screen using their hands and a monitor webcam. If successfully deployed to a large user base, Space Trash could provide researchers with an accessible platform for collecting visuomotor performance data, potentially supporting the development of more accessible detection software.

I. INTRODUCTION

A. Background

Neurological diseases affect millions across the world, with Alzheimer’s Disease (AD) being the most prevalent form of dementia [1]. As the disease progresses, patients experience memory loss, cognitive decline, motor control, and basic quality of life [1]. Although we do not have a cure for Alzheimer’s Disease, we do have the means to slow down these declining factors through several methods if the disease is detected in its early stages. One of

these early detection methods is using the Kinesiological Instrument for Normal and Altered Reaching Movement (KINARM) lab (Figure 1), and its Object Hit and Avoid (OHA) task, testing the participants' spatial awareness, inhibitory control, and reaction time to help detect these early stages as well as help in the data collection in understanding the disease.



Fig. 1: Kinarm End-point Lab

B. Limitations of the KINARM

Though the KINARM is an effective tool for detecting these early stages and collecting data for further research on the disease, there are a few flaws when it comes to the accessibility of the instrument. Currently, there are only around 130 KINARMS distributed across 16 countries including the U.S. [3]. This lack of instruments in many countries causes large gaps in Alzheimer's data collection. The lack of availability also causes researchers that are aiding in Alzheimer's Disease research to have to move to different states depending on institutional access, as well as patients having to spend hundreds of dollars in travel fees just to be tested. The other option would be to have a new instrument created, but depending on what model you need for your research, it will cost you between fifty thousand and two hundred and fifty thousand dollars [4].

C. Space Trash Overview

Space Trash seeks to provide researchers and clinicians with a free, downloadable, gamified version of the Object Hit and Avoid task that can be

deployed on the user's home hardware. Space Trash's goal is to provide researchers a tool for outside of the lab and at home Alzheimer's Disease data collection, filling in those grey areas where the KINARM isn't accessible. With multiple ways of performing this task, and with our first versions being developed with multi-mouse and camera tracking, Space Trash will ensure accessibility to a wide variety of users to ensure there are no hardware limitations.

II. BACKGROUND AND RELATED WORK

A. Alzheimer's Disease and Early Detection

Alzheimer's Disease, named after German psychiatrist Alois Alzheimer, is a disease that causes a degeneration of cells in the brain, with its effects being characterized as a decline in thinking, memory loss, and change in personality [1]. Its causes can be a result of many factors, also known as a multifactorial disease, affecting more than 55 million people across the globe [2].

As it stands, there is no cure for this disease. Despite this, we have discovered treatment that slows down the progression of cognitive decline. One factor that helps with this treatment is catching the disease in its early stages, showing that the treatment is most effective in those stages [5]. Being able to detect these early stages of AD and getting the patients the treatment that they need results in a better quality of life for the patients.

As we developed these early detection methods for AD, we noticed that cognitive decline is not the only factor we can use. Motor impairments in the patients can appear early, allowing us to help diagnose the patient by testing the visuomotor skills through a sequence of tests.

B. KINARM and Object Hit and Avoid

The Kinesiological Instrument for Normal and Altered Reaching Movement (KINARM) is a highly flexible robotic assessment machine that allows researchers to develop their own custom tests. Used

in neuroscience and rehabilitation, the KINARM helps with data collection in coordination of motor actions, solving a plethora of cognitive problems to aid the researcher.

Our project mainly focuses on the Kinarm Endpoint lab and its Object Hit and Avoid task, which tasks users with hitting incoming objects while avoiding distractors. While the user is playing the task, the positional data of where the paddles (the player's hands) are currently at is tracked, tracking where an object or a distractor was hit at the exact position, storing them into a “bucket.” As the task progresses, the production and speed of objects and distractors increase, making the difficulty of the game more challenging, testing exactly where the patient is focusing, as how much of the total area they are covering and with what arm.

OHA uses the positional data of the paddles to measure three core cognitive domains: spatial attention, inhibitory control, and rapid motor decision-making (Figure 2).

Throughout the task, the user has to track multiple objects simultaneously across the screen, testing the brain's visuospatial networks. Early AD can compromise this network, causing the patients to miss targets and fail to notice incoming objects, showing a delayed reaction time.

In early AD, the user can experience a loss of inhibition. The task requires them to stop their reflexes, making them choose what path to follow to hit specific objects. This loss of inhibition results in the user hitting more distractors, as well as having a higher frequency of premature motor responses

As the task increases the speed and production of both objects and distractors, users must adapt their reaction time motor plans to keep up with the task. AD damages the cognitive-motor pathways, causing slower processing and prolonged reaction time.

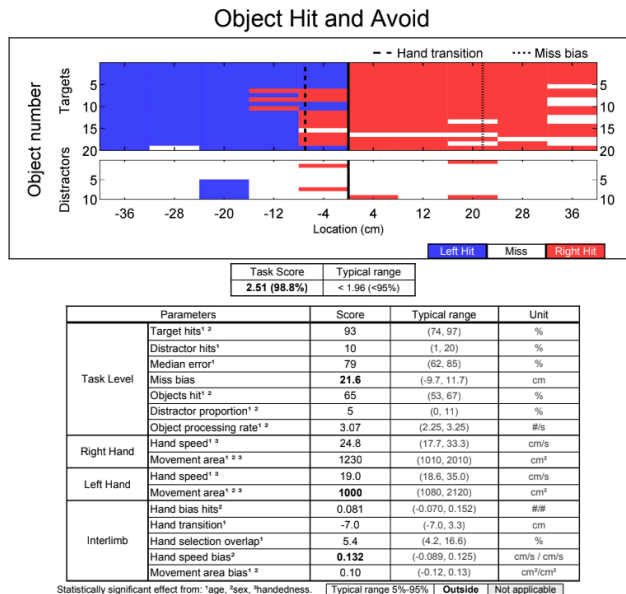


Fig. 2: OHA example data chart

C. Alzheimer's Data Collection Games

Sea Hero Quest (SHQ) was a free mobile game used to collect spatial navigation data to aid AD research [6]. The game was designed to collect navigational ability data, similarly from a large population like Space Trash. With the help of SHQ, data collection for a number of participants skyrocketed, with the amount of participants being tested through the game would ultimately take years to test traditionally. In light of SHQs success, a Minecraft based mod for SHQ was developed [7]. The mod aimed to use the mechanisms of SHQ to create a mod for Minecraft, aiding in early detection of Alzheimer's Disease amongst a younger audience as well as moving to a more mainstream popular game, drawing in more participants to aid in the data collection.

Like SHQ, our game focuses on the data collection side of AD research. Unlike either implementation we've found, Space Trash focuses on using visuomotor skills for the data collection due to working best with recreating the OHA task of the KINARM.

III. METHODOLOGY

A. Hardware and Space Trash

Currently, we are in development of two versions of the Object Hit and Avoid task as a game. Our first version of the OHA task for Space Trash is our camera tracking option, requiring the user to have hardware that cannot only play the game, but also have a webcam either installed or one that can be externally connected to the device. With the use of Mediapipe's tracking software, we will be able to track the positions of the player's hands as they raise them up to the camera, mapping a one to one paddle where the hand is based on where the camera is tracking their hands.

Our second version, as well as the one closer to its first debut, is our two-mice implementation of the task. The user will be asked to use two USB connecting mice (either corded or through dongle), playing a preparation phase that will identify which is the player's right and left arm.

With future validation through testing after the versions are complete, we can determine not only if the recreation of the OHA task was a success, but also record which version was more successful at recreating the data from the KINARM.

B. Different Participating Groups

When it comes to testing our game, we want to make sure that the data we are collecting can be comparable to the results that you would expect from the KINARM. To test this, we plan to have participants play OHA on the KINARM, as well as play all versions of the OHA task for Space Trash. Then, we will be able to look at the raw positional data collected from the KINARM and Space Trash, not only determining whether the game was a success at recreating similar data, but also determining which version of Space Trash was more effective at recreating the data.

We plan to test it against two different types of groups. Our first group will be the general public, since our game will be available to all. We want to

test our game against anyone who might end up playing it, covering a large variety of backgrounds, to see how the game compares to that of the KINARM.

The second group we plan to test Space Trash against is patients who have been diagnosed and/or are currently being treated for Alzheimer's disease, determining whether or not our game can replicate similar data in that caliber

IV. DISCUSSION AND LIMITATIONS

A. Development Decisions

One of the biggest decisions we made for the future of this project was the transition over from our previous work to a game engine called *Godot*. In our previous implementations of the OHA task, we've already run into roadblocks that had caused us to scrap ideas and start from scratch. One thing that we noticed though is that none of the implementations from the previous developers' work used a game engine to help recreate the task.

Considering the current state of Space Trash at the time of the decision, we decided that it would be beneficial to move over to *Godot* to speed up the progress of the game, allowing us to use the game engine's tools to develop the game aspects of the tasks quickly so we can focus on the data collection development side. With our current implementation of Space Trash at the time being camera tracking through Media-pipe, and *Godot* being compatible with Media-pipe, all that needs to be done is translation of the code from Python to GDScript and C#. Another deciding factor when moving over to *Godot* was when we found the multi-mouse plugin for the game engine, allowing us to develop a game where multiple mice can be pointed to different cursors.

While we continue to develop Space Trash, we continue to take into consideration how our game should look and portray to multiple different audiences, with our current focus on making sure that it is portrayed and visually appealing to an elderly audience. One of these decisions was determining

what our color scheme would be throughout the whole game. Since the elderly audience is shown to have an easier time with focusing on bright, warm colors, we will continue to keep that in mind once we are past the testing phase of the game, making sure that we stay within those colors as we prepare it for its accessible package [8].

Another thing to consider with an elderly audience is that having complexity when portraying information about the game, whether it be how to play it or what exactly the game records, having too much information regarding the game can be hard to focus on, especially if there's too much jargon. Keeping that in mind, we plan to develop our game with as simple as possible menus and directions, portraying the information in a simple way while also making sure that everything that needs to be mentioned is [9].

B. Gamers and Space Trash

As we were discussing the groups that we would like to test our game on when it comes to ensuring that the game reflects the data coming from the KINARM, gamers were brought up as a possible problem with multi-mouse. Considering that the online gaming community is already used to playing with a keyboard and mouse, multi-mouse would be a playing field that they would be used to, giving them an edge when it comes to performing the task. We theorize that this difference in skill will alter the data, and hope to get a better understanding of this situation by having gamers participate in playing the game and testing on the KINARM before we release Space Trash to the public.

C. Hardware requirements and problems

We want to keep Space Trash as available as possible, allowing as many people to be given the opportunity to play our game and help aid the research through Alzheimer's data collection. As we continue to replicate the KINARM's requirements for its data collection, we are beginning to run into possible situations where our decisions can lead to how effective our game is at collecting the required amount of data to replicate the results. The KINARM's sampling rate is set at 1000 Hz, allowing

for accurate positional data to be recorded while the user performs the task. With 240 Hz being the standard for high-end racing games, we are concerned about what would happen when testing it on devices that are not equipped with the newer hardware. We will continue to test our sampling rate for Space Trash while we develop our game, finding what rate will allow efficient data collection while also making sure we don't require high-end personal computers to play our game.

Another thing that we noticed while developing Space Trash is how multi-mouse has a different plane when it comes to the user's playing field and their visuals (Figure 3). The KINARM was built to have both the visual and playing field on the same horizontal plane, allowing for better visuomotor skills as well as being easier on older individuals since it's shown that they have a harder time with the plane difference. Multi-mouse will in fact have that planar difference, making it challenging for some users, as well as the possibility of our data being altered [10]. In light of this, we started developing a camera tracking option, allowing users to play on the same plane to gather the same plane data (Figure 3). But since this is only a hypothesis that the plane alters the data, we plan to keep multi-mouse as an option upon release if it shows it replicates the data, allowing us to further study whether a plane difference does, in fact, alter the data. Multi-mouse also adds on to the availability of Space Trash to a variety of systems, since quality of cameras can affect how well the tracking software operates as well as there being cases where systems never had a webcam installed [11].



Fig. 3: Planar difference between versions

D. One-to-one movement and multi-mouse

One of the harder implementations for multi-mouse is making sure that the game creates one-to-one movement from mouse to cursor. When it comes to our camera tracking option, one-to-one movement is already handled, since the camera maps the position of the paddles based on the position of the player's hands. When it comes to multi-mouse, one-to-one movement will have to be handled through settings changes, with our code setting the paddles to a certain speed while also asking the player to ensure that their DPI is set to a certain value.

V. FUTURE WORK

A. Development of multi-mouse

Multi-mouse OHA is currently in the stages of developing the data collection side of the task, with most game aspects of the task already being implemented. Currently, the game is tracking which paddle hits the object, what side the paddle was on, and what specific position the paddle was at when the hit occurred.

We currently still have to develop the distractors of the task, with multi-mouses current implementation looking more like Object Hit (OH) than OHA. We also need to implement the increase of production of the shapes as the game progresses, resulting in more objects to hit and avoid as well as their speed values having the option for a higher speed. But, since these game implementations are easier to implement through our game engine, we will first focus on the data collection side of the task since there are a lot of aspects that need to be taken into account while developing. When it comes to the data collection side of the test that still needs to be developed, we still need to develop a way to pull that positional data of the players paddles in a certain frequency so that it can be compared to the KINARM's results, as well as work out a bug that the game is currently showing where one paddle tracks left and right hits while the other only tracks hits from the side it first hits in.



Fig. 4: Multi-mouse Space Trash

B. Development of camera tracking

We are currently still in the process of transferring our past implementation of camera tracking to the current game engine that we plan to continue using while creating this game. In our past implementation, we had most of the game aspects done, with only having the distractors and increased production of objects needing to be developed, similarly with where multi-mouse is now.

When it comes to the data collection side of camera tracking, we're also experimenting with exactly how frequently we want to store the positional data. We've also noticed that there is a bug when it comes to the hit detection software, with some hits counting towards two.

Our current version of camera tracking Space Trash for our game engine is still in its early stages of development, with fluid hand tracking still needing to be developed.

Since the development of both versions are progressing smoothly, we are moving closer and closer to our first test of Space Trash. With the work that has been done and it being actively worked on, we hope to have our first test with at least multi-mouse by Fall of 2026

VII. REFERENCES

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